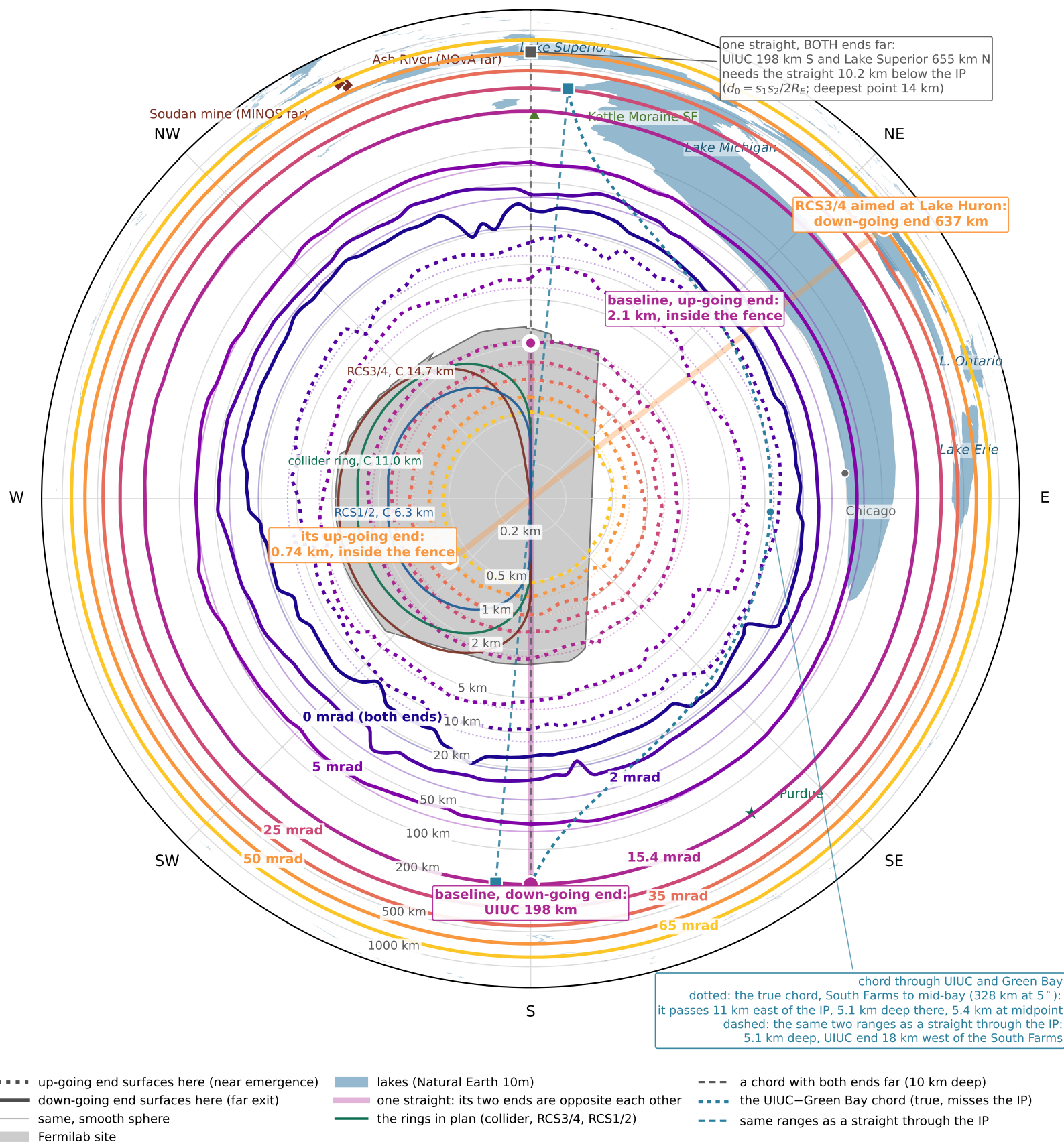


Where the two ends of one straight surface, tilt by tilt  
 35 m-deep straight; one colour per tilt – dotted: the end that goes up, solid: the end that goes down; at 0 mrad they are the same curve  
 N



Bold curves: USGS NED 10 m terrain inside 6 km and GEBCO 2020 beyond, along 144 bearings (0.1–1000 km), Catmull-Rom smoothed; exits over the Great Lakes taken at the water surface. Faint circles: smooth sphere,  
 $s_{near} = R_E(-\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx d_0/\theta$  and  $s_{far} = R_E(\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx 2R_E\theta$ . Log range; rings in km. The fence is 0.9–4.9 km out, so the dotted curves show which tilts emerge on site.